Pre-Lab 4: Transmission Using DSB-SC Modulation and USRP Noise Measurement

Objectives

A) To design transmitter and receiver for DSB-SC modulation.

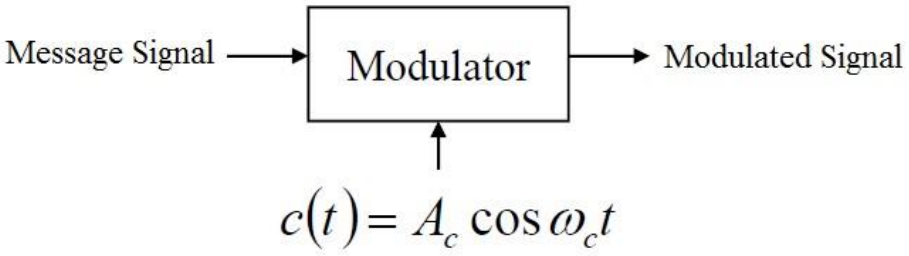
B) To calculate power efficiency (power of message/ total power)

C) To calculate bandwidth of the message signal.

D) To calculate bandwidth of modulated signal.

E) Introduction to USRP and investigate the noise of USRP

In communication, modulation is the process of varying one or more properties of a highfrequency periodic waveform, called the carrier signal, with a modulating signal which typically contains information to be transmitted, called the baseband (message signal). A modulator is used to modify a carrier signal's parameter (frequency, amplitude, phase, etc.) in accordance to the baseband signal.

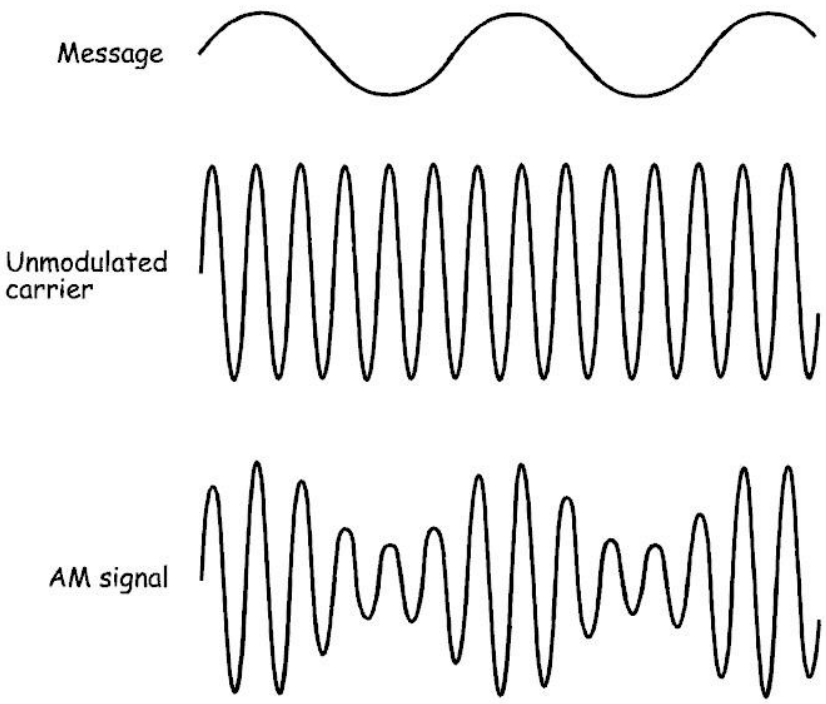


Modulation is necessary in the transmission of data, such as an audio signal, over a long range. Baseband signals are inherently limited in range because their frequency range is typically in a lower range and their power makes the signal unsuitable for transmission. Modulation gives several advantages, such as reducing the size of transmitting antenna, better usages of limited bandwidth, and much more.

I. Theory

In an amplitude modulation (AM) communications system, the message signal is used to vary the amplitude of a pure cosine wave called the carrier. The carrier usually has a frequency (f_c

) that is much higher than the message's frequency. Figure below shows a simple message signal and an unmodulated carrier. It also shows the result of amplitude modulating the carrier with the message. Notice that the modulated carrier's amplitude varies above and below its unmodulated amplitude.



Double Sideband Supressed Carrier (DSB-SC) Modulation:

Let's denote the baseband message signal with $m\,t(\,)$ and the carrier signal with $c\,t(\,)=A_c\cos 2(\,\square f\,t_c)$ where A_c is amplitude and f_c is carrier frequency. The DSB-SC modulated signal is given as

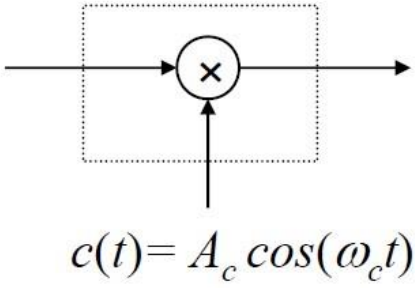
$$u\,t(\,)=m\,t\,c\,t(\,)\,(\,)=m\,t\,A(\,)_{\,c}\cos 2(\,\square f\,t_c) \qquad (1)$$

The spectrum of the modulated signal is

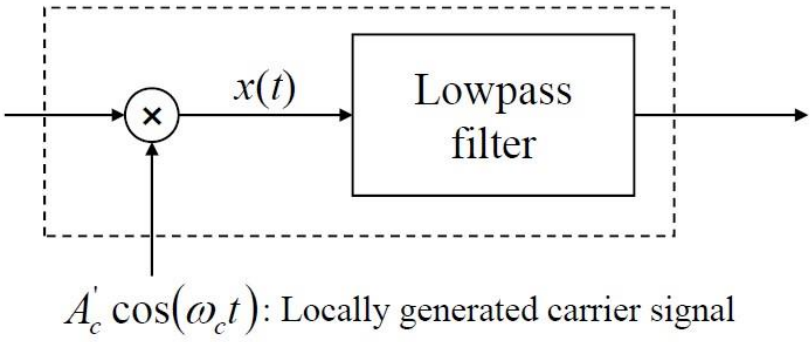
$$U f(\,)=\frac{}{}\;A^c M f(\,-\,\frac{}{}\;+f_c)\quad A^c M f(\,+f_c)$$

2 2 where $M(f)$ is spectrum of message signal $m(t)$, and $U(f)$ is shifted versions of $M(f)$ into f_c and $-f_c$ frequencies.

Modulation: DSC-SB modulation is multiplication of message signal $m(t)$ and $c(t)$.



Demodulation: For DSB-SC, we use coherent demodulation as shown in figure below.



II. Pre-Lab 4

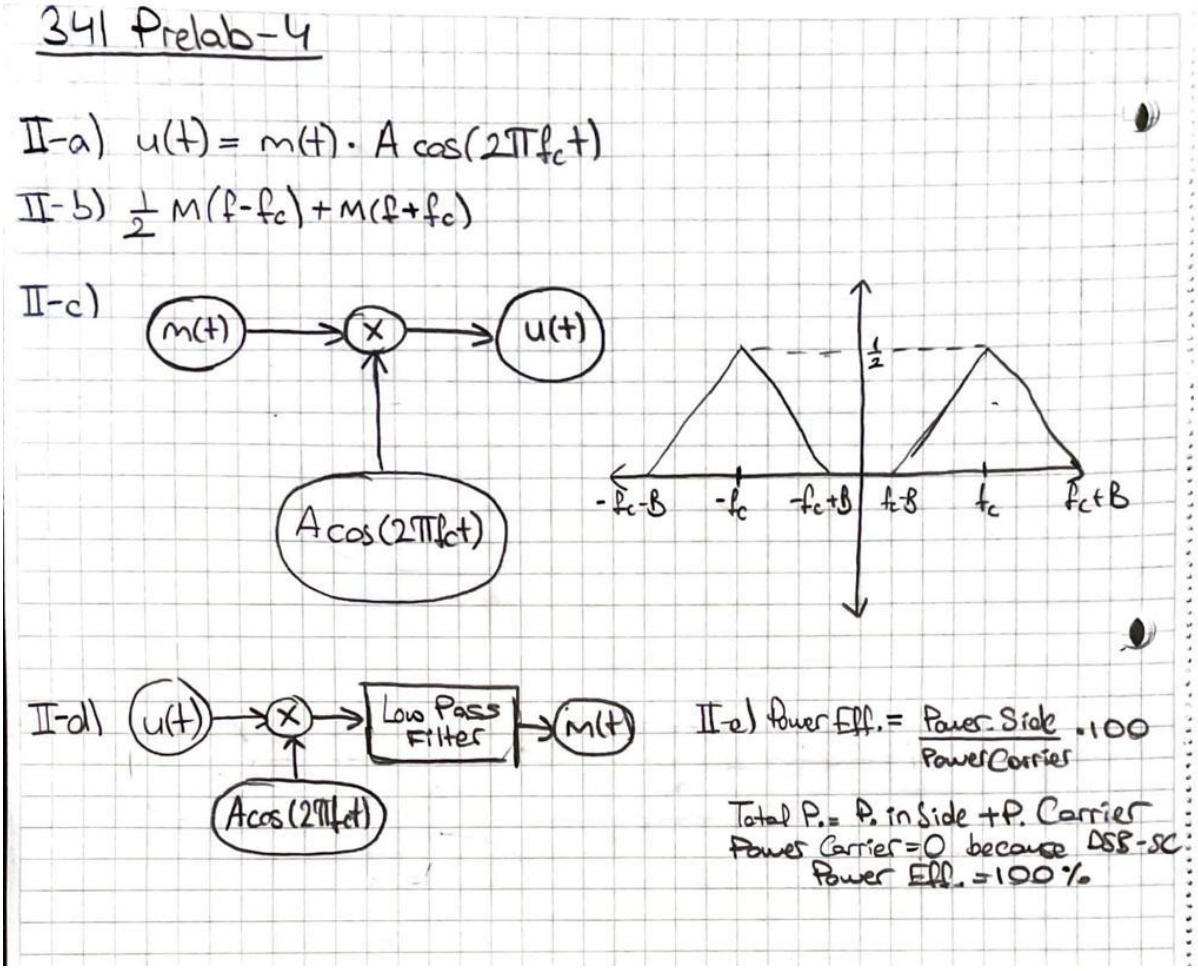
In the pre-lab part you need to answer to the following questions and also make the LabVIEW block diagram and bring it to the lab. **II. a. Questions**

Let's denote the baseband message with $m(t)$ and the carrier signal with $c(t) = A_c \cos 2(\square f t_c)$ where A_c is amplitude and f_c is carrier frequency.

a) Write down the DSB-SC modulated signal $u(t)$.
b) Write down and plot The spectrum of the modulated signal.
c) Plot a block diagram which gives the modulated signal $u(t)$ as output.

d) Plot the block diagram for coherent demodulation.

e) Determine the power efficiency of DSB-SC modulation.



Power Spectral Measurements and Noise of USRP

In this part of the lab, we investigate the power spectrum of a Gaussian white noise and study the effect of filtering. In the last part of the lab, we turn on the USRP and calculate the noise power of electrical components inside the USRP and apply a filter to decrease it.

Theory

The Power Spectrum VI, which is closely related to the FFT, calculates the harmonic power in a signal. The power spectrum, $S_x(f)$, of a time domain signal, $x(t)$, is defined using the following equation:

$$S_x(f) = |X(f)|^2 \quad (5)$$

The discrete implementation of the power spectrum, $\boldsymbol{S}_x$, is given by the following equation:

$$S_x = \frac{|\mathbf{F}(x)|^2}{n^2} \quad (6)$$

Unlike the FFT, power spectrum results are always real. Furthermore, the power spectrum uses the single-sided format because this format eliminates redundancy while retaining the complete power spectrum information and hence uses less memory.

Filters alter or remove unwanted frequencies from your signal. Depending on the frequency range that they either pass or attenuate, they can be classified into the following types:

- A **lowpass** filter passes low frequencies, but attenuates high frequencies.
- A **highpass** filter passes high frequencies, but attenuates low frequencies.

• A **bandpass** filter passes a certain band of frequencies.

• A **bandstop** filter attenuates a certain band of frequencies.

III. Pre-LabVIEW Programming (Power Spectrum, Noise, and Filters)

To graphically display the results of power spectrum and noise follow the below procedure:

1. Launch a new blank VI.
2. Create a numeric control and name it "t (duration of noise)".
3. Add a **Gaussian White Noise VI (Signal Processing => Signal Generation)**. Create the necessary inputs.
4. Calculate □ □*t t n*/ by dividing duration of noise with number of samples. Add a bundle and display the noise using graph.
5. In order to display the power spectrum of the noise, add **Auto Power Spectrum VI (Signal Processing => Spectra)**. Connect the noise signal and **dt** to the inputs. By using a bundle and **df** output of power spectrum, display the spectrum of noise.
6. Next step is to use a filter to observe the power spectrum of filtered noise. Hence add a **Butterworth Filter (Signal Processing => Filters)**. Create controls for filter type, high and low cut off frequency. For sampling frequency input, f_s □ □1/ t , apply the inverse of dt to the filter. Connect the noise signal to X input.
7. Display the power spectrum of filtered noise same as step 5.
8. Run the VI and set t=10, samples=10000, amplitude=1, filter type=lowpass.
9. Increase the low cut off frequency slightly and observe the power spectrum. The low frequency part of the signal should appears.
10. Manipulate filter type, low and high cut off frequency while observing the power spectrum of filtered noise.

11. Save the VI.

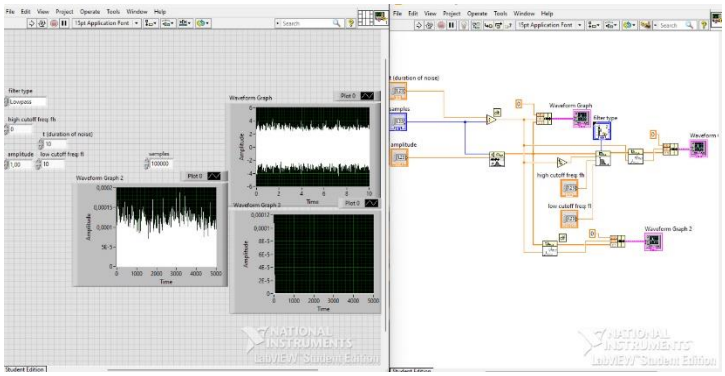
III.a. Questions:

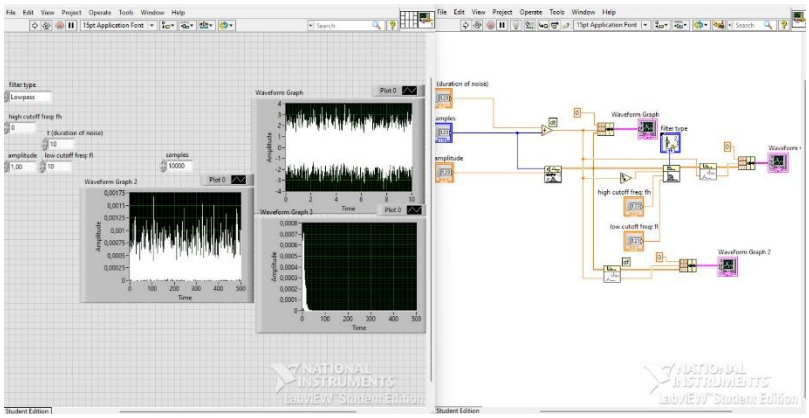
1. Calculate the number of samples in 1 sec duration of noise.

For the number of samples in 1 sec of duration of noise we will use the Formula "Number of samples = Sampling Rate * Time". First, we should find the Sampling rate so 10000 = Sampling rate * 10, Sampling Rate = 1000 Hz. We want to set at t=1 so number of samples should be equal to the 1000Hz

2. Increase the number of samples and explain why the x-axis values of power spectrums are increasing? As you increase the number of samples more frequencies can be observed in graph. Explain for all of the frequencies the power of noise is nonzero?

When we increase the number of samples, the x-axis values of power spectrums increase because the frequency resolution also increases. Because we keep the time constant and we increase the number of samples, this happens to be an increase in frequency resolution. All the frequencies in the power of noise are nonzero because there is a white noise with a constant power density in the frequency spectrum.





3. Calculate the power of the noise by taking integral from power spectrum of noise. For integration use **Numeric Integration VI (Mathematics => Integ & Diff).**

